COMP9517: Computer Vision

2026 Term 1

Group Project Specification

Maximum Marks Achievable: 40

The group project is **worth 40% of the total course mark**.

Project work is in Weeks 6-10 with deliverables due in Week 10.
Deadline for submission is Friday 24 April 2026 18:00 (Sydney time).
Instructions for online submission will be posted closer to the deadline.
Refer to the separate marking criteria for more detailed information on marking.

Introduction

The goal of the group project is to work together with peers in a team of 5 students to solve a computer vision problem and present solutions and results in both oral and written form.

Group members can meet with their tutors once per week in Weeks 6-10 during the usual tutorial session hour to discuss progress and get feedback.

Description

Reliable and automated segmentation of crop is a fundamental visual task supporting a multitude of agricultural practices, including the monitoring of crop growth, assessment of pest conditions, classification between the crop of interest and weeds, and the distinction between healthy and diseased plant tissues. Computer vision methods can help to precisely segment crop from field images.

Task Statement

The goal of this group project is to develop and compare different computer vision methods for segmenting wheat crop in images under diverse and dynamic conditions.

Public Dataset

The public dataset to be used in this group project is EWS (Eschikon Wheat Segmentation) and is available from EWS-Dataset.zip (download the "EWS-Dataset.zip" file; see reference and link at the end of this document). It contains 190 RGB images of 350 x 350 pixels with manual annotation in the form of binary masks for plants and soil, respectively.

Method Development

Many traditional, machine learning, and deep learning-based computer vision methods could be used for this task. You are challenged to use concepts taught in the course and other techniques from literature to develop your own methods and test their performance.

You are not expected to implement everything from scratch. You may adapt existing implementations, modify known methods, or apply more recent techniques that were not covered in the lectures. However, you must clearly cite all sources, including papers, software libraries, and code repositories that you use.

Basic Requirements

Each group must develop and evaluate at least three different segmentation methods. These methods must be meaningfully different approaches, not minor parameter changes to the same method. In addition, thresholding methods would not be counted, as it was already explored in Lab 4.

Comprehensive Method Development

If you are aiming for higher marks (33-36), we expect to see greater diversity and thoughtful comparison of methods. For example, as a group, you could explore methods from at least two of the following categories:

- Advanced segmentation methods (Week 5 lectures), comparing different approaches.
- Machine learning-based methods, comparing different handcrafted feature descriptors and classification methods.
- Deep learning-based methods, comparing different deep neural network architectures and data augmentation methods.

Advanced Method Development

If you are aiming for even higher marks (37 and above), we expect to see some original ideas, deeper analysis and research skills. For example, you could consider the following:

- Simulate realistic image distortions such as noise, blur, low brightness/contrast, or partial occlusion and evaluate how your methods perform with such images. Investigate techniques that improve robustness, such as improved feature design, deep learning model design, training strategies and postprocessing methods.
- Conduct in-depth performance and failure analysis, especially when less precise ground truth labels or fewer training images are used during training. Use insights from your analysis to improve your methods, for example through model adjustments, loss function modifications, or improved training procedures.
- You are also encouraged to explore additional original ideas that enhance your methods and results, with clear justification for why these ideas are worth investigating. You may discuss your ideas with the tutors during the weekly tutorial hours.

Training and Testing

The given dataset has already been split into training, validation, and test subsets. It is very important to never mix them up or cause data leakage. Whether your methods require training (supervised methods) or not (unsupervised methods), they will probably have (hyper)parameters that you need to fine-tune to get optimal performance. Use the training and validation sets for this and use the test set only for final performance evaluation.

To assess the performance of your methods, compare the predicted mask with the corresponding annotated mask (ground truth) using appropriate metrics. Specifically, report metrics such as the precision, recall, F1-score and intersection over Union (IoU). Furthermore, compare the training and testing times of the different methods.

Results and Discussion

Show the quantitative results in your video presentation and written report (see deliverables below). Also show representative examples of successful results as well as examples where your methods failed (no method generally yields perfect results). Give some explanation why you believe your methods failed in these cases. Also, if one of your methods clearly works better than the other(s), discuss possible reasons why. Finally, discuss some potential directions for future research to further improve performance.

Practicalities

The dataset is less than 150 MB in total, so method training and testing should be feasible on a modern desktop or laptop computer. If more computing resources are needed, you could consider using the free version of Google Colab. Otherwise, you are free to use only a subset of the data, for example 75% or even just 50%. Of course, you can expect the performance of your methods to go down accordingly, but as long as you clearly report your approach, this will not negatively impact your project mark.

Deliverables

The deliverables of the group project are 1) a video presentation, 2) a written report, and 3) the code. The deliverables are to be submitted by only one member of the group, on behalf of the whole group (we do not accept submissions from multiple group members).

Video

Each group must prepare a video presentation of at most 10 minutes (any content beyond that will be ignored). The presentation must start with an introduction of the problem and then explain the used methods, show the obtained results, and discuss these results as well as ideas for future improvements. This part of the presentation should be approximately 8 minutes long, using PowerPoint slides to support the narrative.

Following this part, the presentation must include a demonstration of the methods in action, in about 2 minutes. Of course, some methods may take a long time to compute, so you may record a live demo and then edit it to stay within time. The objective of this demonstration is to briefly show how the code works and what outputs it produces. If your group develops a large number of methods, you may only be able to demonstrate a subset of them.

All group members must present (points will be deducted if this is not the case). It is up to you to decide who presents what (introduction, methods, results, discussion, demonstration). In order for us to verify that all group members are indeed presenting, each student presenting their part must be visible in a corner of the presentation (live recording, not a static head shot), and when they start presenting, they must mention their name.

Overlaying a webcam recording can be done using either the video recording functionality of PowerPoint itself (see for example this YouTube tutorial) or using other recording software such as OBS Studio, DaVinci Resolve, Adobe Premiere, and many others. It is up to you (depending on your preference and experience) which software to use, as long as the final video satisfies the requirements mentioned above and below.

The entire presentation must be in the form of a video (720p or 1080p MP4 format). **To avoid storage problems for this course, the video upload limit is 100 MB per group, which should be more than enough for this type of presentation.** If your video file is initially larger, use tools like HandBrake to re-encode with higher compression. For example, in Handbrake, choose the "Very Fast 720p30" Preset and then go to the Video tab to lower the Quality as needed. That way, even a 1-hour video presentation can be compressed to under 100 MB without significant loss of quality. Minor compression artifacts will not affect your mark.

The video presentations will be marked offline (there will be no live presentations). If the markers have any questions, they may contact the group members by email for clarification.

Report

Each group must also submit a written report (in 2-column IEEE format; maximum 10 pages of main text including all diagrams, figures, tables, but excluding references; you may add any number of additional pages for references only).

The report must be submitted as a PDF file and include:

1. Introduction: Present a brief introduction including the relevance and your understanding of the task and the dataset.
2. Literature Review: Review relevant techniques in literature, along with any necessary background to understand the methods you selected.
3. Methods: Motivate and explain the methods you implemented, using relevant references and theories where necessary.
4. Experimental Results: Explain the experimental setup you used to test the performance of the developed methods and the results you obtained.

5. Discussion: Provide a discussion of the experimental results and findings, comparing the different methods you developed.
6. Conclusion: Summarise what worked and what did not work, any limitations of the developed methods, and recommend future work.
7. References: List the literature references and other resources used in your work. The references section does not count toward the 10-page limit.

The upload limit for the report is 10 MB per group. If your report is initially larger than that, you may need to resize images or increase the compression.

Code

The source code of the developed software must be submitted as a ZIP file. The file should mainly contain your own code. Libraries or code obtained from other sources should be clearly described in a README file. Include proper documentation about how to run the code, use inline comments, and make the code well structured.

The upload limit for the source code is 25 MB per group. To stay within this limit, do not include trained models, input images, or result images.

Student Contributions

As a group, you are free in how you divide the work among the group members, but all group members must contribute roughly equally to the method development, coding, making the video, and writing the report. For example, it is unacceptable if some group members only prepare the video or report without also contributing to the methods and code.

An online survey will be held in Week 11 (after the project deliverables submission deadline). It will allow students to anonymously evaluate the relative contributions of their group members to the project. The results will be reported only to the LIC and the Course Administrators, who at their discretion may moderate the final project mark for individual students if there is sufficient evidence that they contributed substantially less than the other group members. Submission instructions will be posted on WebCMS3 later.

Dataset

The dataset to be used in the group project is the following:

Zenkl et al. Outdoor Plant Segmentation with Deep Learning for high-throughput Field Phenotyping on a diverse Wheat Dataset. Frontiers in Plan Science, 12:774068. https://doi.org/10.3389/fpls.2021.774068

EWS Dataset Download Link (use the "EWS-Dataset.zip" file):

https://www.research-collection.ethz.ch/entities/researchdata/165d22fc-6b0f-4fc3-a441-20d8bdc50a70

References

The following papers may be helpful in getting a deeper understanding of the problem and exploring potential methods:

Zenkl R, Timofte R, Kirchgessner N, Roth L, Hund A, Van Gool L, Walter A and Aasen H (2022) Outdoor Plant Segmentation With Deep Learning for High-Throughput Field Phenotyping on a Diverse Wheat Dataset. Front. Plant Sci. 12:774068. doi: 10.3389/fpls.2021.774068
https://doi.org/10.3389/fpls.2021.774068

Songliang Cao, Binghui Xu, Wei Zhou, Letian Zhou, Jiafei Zhang, Yuhui Zheng, Weijuan Hu, Zhiguo Han, Hao Lu, The blessing of Depth Anything: An almost unsupervised approach to crop segmentation with depth-informed pseudo labeling, Plant Phenomics, Volume 7, Issue 1, 2025, 100005, ISSN 2643-6515.
https://doi.org/10.1016/j.plaphe.2025.100005

Zhang *et al*., "DepthCropSeg++: Scaling a Crop Segmentation Foundation Model With Depth-Labeled Data," in *IEEE Journal of Selected Topics in Signal Processing*, doi: 10.1109/JSTSP.2026.3654362.
https://ieeexplore.ieee.org/document/11353458

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